Multimodal Mamba Fusion for Noise-Robust EEG Biometric Identification

Ly Ngoc Vu*†, Huynh Cong Bang*† and Nguyen Minh Hanh*‡
* Faculty of Information Technology, Industrial University of Ho Chi Minh City, Vietnam
E-mail: dezzhuge@gmail.com, bang_01036047@iuh.edu.vn, hanh_01036046@iuh.edu.vn
†Equal contribution (co-first authors). ‡Corresponding author.

***Abstract*—EEG biometric identification is brittle under deployment noise and unseen-subject conditions. Extending our two-stage WaveNet+ResNet pipeline, we propose a multimodal Mamba framework: a Mamba-augmented denoiser, a parallel spectrogram branch with frequency- and time-axis Mamba sweeps, and a gated self-attention fusion module with auxiliary branch losses. On the PhysioNet AEP dataset under the AEP-Hybrid protocol (16 known + 4 held-out impostor subjects, three seeds, inheriting the partition of [1]), the full model (V4) attains the best Precision@1 on all three noise families (82.4% Gaussian, 87.7% power-line, 83.8% EMG) with Stage-I SI-SNR 12.15/32.38/13.83 dB. Held-out-subject AUROC ≈ 0.50 motivates calibration as future work.**

***Index Terms*—EEG biometrics, denoising autoencoder, state-space models, Mamba, multimodal fusion, gated self-attention fusion, metric learning.**

I. Introduction

The push toward unobtrusive authentication for wearables, brain-computer interfaces, and cyber-physical systems has revived interest in EEG-based biometrics: the signal is internal, hard to spoof, and rich in subject-specific micro-patterns [2], [3], [4]. Concretely, we treat EEG biometrics as the following task: given a 4-channel, $T = 800$-sample EEG window from an enrolled subject, possibly corrupted by deployment noise, output a unit-norm 128-d biometric embedding such that (a) the closest enrolled identity in cosine distance is the correct subject (closed-set retrieval), and (b) an unseen subject can in principle be flagged via a distance threshold (open-set verification). Two practical challenges persist: real recordings are corrupted by power-line interference, muscle artifacts, and broadband noise [5], and deployable systems must generalize to subjects that were never seen during training [6], [7].

Our prior two-stage pipeline [1] decouples a WaveNet denoiser from a ResNet metric-learning embedder but operates on a single view of the signal with a fixed convolutional receptive field. This paper grounds three contributions in concrete observations on that design. **(i)** The denoiser's receptive field is fixed by depth and dilation, so we insert a *selective state-space* (Mamba) [8], [9] block at the midpoint of the WaveNet stack for content-aware mixing in linear time. **(ii)** The raw waveform hides rhythmic line-noise structure, so we add a *spectrogram branch* that re-encodes the denoised signal into a 2-D time-frequency map with frequency- and time-axis Mamba sweeps. **(iii)** A single embedding view risks branch-specific bias, so we fuse the two embeddings with a *gated self-attention fusion* module, jointly trained with auxiliary metric losses on each branch.

We evaluate four progressively richer architectures on the PhysioNet Auditory-Evoked-Potential EEG dataset [5]: V1 is the WaveNet-only denoiser baseline of [1]; V2 inserts a Mamba block at the WaveNet midpoint; V3 is a single-seed H100 quick run with a tuned Mamba preset, reported as a directional reference; V4 is the full multimodal proposal. Under the AEP-Hybrid protocol (Sec. IV), V4 delivers the highest mean closed-set Precision@1 on every noise family in our benchmark and reaches up to 32.4 dB SI-SNR on power-line noise.

II. Related Work

EEG biometrics. Early EEG biometric systems relied on hand-crafted spectral features [2]. Deep models then took over, including attention-recurrent architectures [4], triplet-loss embeddings on raw signals [3], and Siamese networks for cross-session matching [2]. Compact convolutional networks such as EEGNet [10] and deep ConvNets for EEG decoding [11] provide widely used backbones, but they are generally trained for closed-set classification and do not explicitly target open-set verification.

Denoising and signal enhancement. WaveNet-style dilated convolutions are a strong, well-studied tool for raw-waveform denoising [12], [13]. Our prior work [1] adopts this denoiser with an SI-SNR objective; the current paper investigates whether replacing part of the convolutional pathway with a selective state-space block produces a qualitatively cleaner signal.

State-space models. Structured state-space sequence models (S4) [9] and the more recent Mamba [8] approximate long-range temporal dependencies in linear time, with content-dependent gating. They are attractive for biomedical time-series where Transformer attention is costly [14] and where physiological non-stationarity makes hand-tuned filters fragile. EEG-specific Mamba variants have begun to appear for cross-subject decoding [15]; to our knowledge, ours is the first study that combines a Mamba-augmented denoiser with a Mamba spectrogram branch and gated self-attention fusion for EEG biometric identification with open-set verification analysis.

Metric learning and multimodal fusion. Angular margin and pair-weighted losses (ArcFace [6], multi-similarity [7], contrastive [16], triplet [17]) are standard tools for open-set verification. Audio spectrogram transformers [18] demonstrate that converting a 1-D acoustic signal into a 2-D time-frequency map and applying sequence models on top is effective; we adopt the same intuition with Mamba sweeps along both spectrogram axes and use a gated self-attention fusion to combine the two views, instead of a simpler concatenation or score-level fusion.

III. Method

A. Overall Pipeline

The pipeline keeps the two-stage decomposition introduced in [1] but enriches each stage. Stage I is responsible for removing deployment noise from the raw waveform, and Stage II is responsible for mapping the cleaned signal to a 128-d biometric embedding. Stage I is a WaveNet+Mamba denoiser trained with a scale-invariant signal-to-noise ratio (SI-SNR) loss; Stage II is a multimodal embedder that consumes the denoised waveform together with a short-time Fourier transform (STFT)-based spectrogram and emits a 128-d L2-normalized biometric embedding (the L2-normalization layer is what produces the unit-norm vectors used downstream). The two stages are trained sequentially: Stage I for 30 epochs and then frozen as the upstream module for Stage II.

Fig. 1 details the V4 multimodal architecture, which augments the second stage with a spectrogram branch and gated self-attention fusion.

B. Stage I: noise-conditioned waveform denoiser

The denoiser ingests a 4-channel raw EEG epoch $x_{\text{noisy}} \in \mathbb{R}^{4\times 800}$ and emits a clean estimate $\hat{x} = \mathcal{D}(x_{\text{noisy}})$. The WaveNet backbone contributes multiscale dilated convolutions for raw-waveform modeling, while the inserted Mamba block contributes selective, content-aware long-range mixing in $\mathcal{O}(L)$ operations. We retain the dilated-convolution backbone of [12], [13]: three blocks of four gated dilated layers with dilation 2^i for $i \in \{0,1,2,3\}$. At the midpoint of the stack we insert a single *Mamba block* that performs content-dependent state-space mixing over the temporal axis with $d_{\text{model}} = 64$, $d_{\text{state}} = 16$, kernel 4, and expand factor 2. A pre-LayerNorm and a residual connection wrap the block. Architecturally, this swaps a single fixed-receptive-field convolutional layer for a selective state-space layer that re-weights information across the entire epoch in $\mathcal{O}(L)$ operations.

We synthesize paired training data, i.e. (noisy input, clean target) EEG pairs, by mixing clean EEG with three noise families: Gaussian, 50 Hz power-line, and broadband EMG-like (20–80 Hz) at controlled SNR levels. The training objective is the SI-SNR loss, a scale-invariant reconstruction-quality indicator standard in source separation [13], used here so that Stage I is robust to amplitude rescaling between the clean target and the reconstructed estimate. The denoiser is trained for 30 epochs:

$$\mathcal{L}_{\text{SI-SNR}}(\hat{x}, x_{\text{clean}}) = -10\,\log_{10}\frac{\|\alpha x_{\text{clean}}\|^2}{\|\hat{x}-\alpha x_{\text{clean}}\|^2}, \quad \alpha = \frac{\langle \hat{x}, x_{\text{clean}}\rangle}{\|x_{\text{clean}}\|^2}. \tag{1}$$

C. Stage II: dual-view biometric embedder

EEG branch. The denoised epoch is reshaped from $(4,800)$ to $(4,25,32)$ and passed through a ResNet-18 or ResNet-34 backbone [19] with a 3×3 stride-1 stem and no max-pool to preserve the small spatial structure, followed by a projection head Linear$(\cdot,256)$ $\to$ ReLU $\to$ Dropout(0.1) $\to$ Linear$(256,128)$ $\to$ BatchNorm and L2 normalization, producing the unit-norm waveform embedding $e_{\text{eeg}} \in \mathbb{R}^{128}$.

Spectrogram branch. We compute a short-time Fourier transform of the denoised signal with $n_{\text{fft}} = 128$ and hop length 64, yielding a 4-channel time-frequency tensor $S \in \mathbb{R}^{4\times 65\times 13}$. A two-layer convolutional stem (4→32→64 channels with 3×3 kernels and BatchNorm–ReLU) projects S into a 64-d feature map. Following the audio-spectrogram-transformer intuition [18] of decomposing a 2-D map into two 1-D sequence problems, we then apply two Mamba sweeps with a shared hidden width of 64: *frequency-axis* (sequence is the frequency axis with the time axis flattened into the batch) and *time-axis* (sequence is the time axis with frequency flattened); the first sweep models per-band rhythmic structure, the second models its temporal evolution. After mean-pooling over the spatial dimensions, an MLP head produces the unit-norm spectrogram embedding $e_{\text{spec}} \in \mathbb{R}^{128}$ via a final L2-normalization layer that is what makes the embedding unit-norm.

Gated self-attention fusion. The two unit-norm vectors are stacked into a length-2 token sequence $T = [e_{\text{eeg}}; e_{\text{spec}}] \in \mathbb{R}^{2\times 128}$ and passed through a four-head self-attention block (shared $Q/K/V$ derived from T; referred to as a cross-attention module in [1]). The attended tokens $\tilde{T} = [\tilde{e}_{\text{eeg}}; \tilde{e}_{\text{spec}}]$ are mixed by a learned gate $g = \sigma(W_g[e_{\text{eeg}}; e_{\text{spec}}]) \in (0,1)^{128}$ computed from the pre-attention embeddings, where σ is the elementwise sigmoid and $W_g \in \mathbb{R}^{128\times 256}$ is a learned linear projection acting as a soft per-coordinate selector between the two views,

$$e_{\text{mix}} = g \odot \tilde{e}_{\text{eeg}} + (1-g) \odot \tilde{e}_{\text{spec}}, \tag{2}$$

followed by a residual MLP and L2 normalization to give the fused biometric embedding $f \in \mathbb{R}^{128}$ used at inference.

Joint training. Stage II is trained for 30 epochs with one of two metric losses on f: ArcFace [6] or multi-similarity [7], drawn from [20]. To prevent the fusion module from collapsing onto one branch, we add weighted auxiliary metric losses on each branch:

$$\mathcal{L}_{\text{II}} = \mathcal{L}_{\text{metric}}(f) + \lambda_e \mathcal{L}_{\text{metric}}(e_{\text{eeg}}) + \lambda_s \mathcal{L}_{\text{metric}}(e_{\text{spec}}). \tag{3}$$

Numerical values for λ_e, λ_s, and the `MPerClassSampler` parameter m are fixed across all experiments and reported in Sec. IV; the chosen operating point lies on the plateau of single-seed sensitivity sweeps that we defer to the journal version.

IV. Experimental Setup

Dataset. We use the publicly available PhysioNet *Auditory-Evoked-Potential EEG-Biometric* corpus [5], recording 20 subjects across 12 sessions of approximately two minutes each under both resting-state and auditory-stimulation conditions, with in-ear and bone-conduction headphones. Auditory evoked potentials are the cortical response to a controlled acoustic stimulus; their latency- and amplitude-domain morphology can carry subject-specific structure that originates in individual differences of cortical anatomy and auditory processing pathways, which makes them attractive as an internal, replay-resistant biometric cue. Recordings are segmented into fixed-length 4-channel epochs of $T = 800$ samples.

Evaluation protocol. We inherit the subject partitioning and open-set scoring rule of [1] and refer to it as the *AEP-Hybrid protocol* for brevity below. Concretely, 4 of the 20 subjects ($\mathcal{S}_{\text{holdout}} = \{2, 5, 7, 12\}$) are strictly held out as an unknown-impostor pool, and the remaining 16 known subjects are split per-subject into train/validation/test along sessions and epochs. Two evaluation tasks share this split. *(i) Closed-set retrieval* reports P@1 and P@5 on the test split of the 16 known subjects (session- and epoch-disjoint, identity space seen at training). *(ii) Held-out-subject open-set scoring* forms a per-subject prototype as the mean of training-set fused embeddings, scores each held-out epoch by its minimum Euclidean distance to any prototype, and fixes the operating threshold at the 95th percentile of the same min-distance on validation. Held-out-subject AUROC under this scoring stays near chance (≈ 0.50) and is reported in the discussion; the AUROC and EER columns in Table III are closed-set pairwise verification scores on the same test pool, characterizing within-subject session variability rather than held-out-subject novelty. We repeat each configuration with three random seeds (PyTorch, NumPy, and Python `random` seeds $\{0, 1, 2\}$) and report mean$\pm$standard deviation.

Noise families. We synthesize three noise types at deployment: white Gaussian, 50 Hz sinusoidal power-line, and broadband 20–80 Hz EMG-like noise.

Hyperparameters. The auxiliary-loss weights are fixed at $\lambda_e = 0.3$ and $\lambda_s = 0.2$ and the class-balanced sampler parameter at $m = 4$. These operating points lie on the plateau of single-seed sensitivity sweeps over $\lambda_e \times \lambda_s$ and $m \in \{2, 4, 8, 16\}$ on the R34+ArcFace + power-line configuration; the full grids are deferred to the journal version, while the headline three-seed runs in Tables I and III use the fixed operating point above.

Architectures. We compare four variants under identical data, splits, seeds, and training schedules:

- **V1**: WaveNet denoiser only (no Mamba) + EEG-only metric embedder. This is the prior pipeline of [1] re-evaluated under the AEP-Hybrid protocol.
- **V2**: V1 with the midpoint Mamba block in the denoiser.
- **V3**: V2 with a tuned Mamba preset, run as a directional, single-seed, different-hardware quick run (NVIDIA H100 with `torch.compile`); not used for the controlled three-seed claims.
- **V4**: full multimodal model proposed in this paper: WaveNet+Mamba denoiser, ResNet EEG branch, Mamba spectrogram branch, and gated self-attention fusion with auxiliary branch losses.

Implementation. All three-seed V1, V2, and V4 experiments run on a single NVIDIA RTX 5090 with batch size 256, 8 data-loader workers, mixed precision, and the `denoised` spectrogram source for V4. The V4 model totals 22.8 M parameters with end-to-end inference latency of 186–197 ms per 4 s epoch on an RTX 5090 (ResNet-34+ArcFace head), well below the 4 s epoch duration and therefore real-time-compatible; the V1 baseline reaches $\sim$20 ms on an RTX 3060 with comparable parameter count ($\sim$23.4 M) [1], so the added Mamba blocks, spectrogram branch, and gated fusion module account for the latency delta on top of the V1 backbone. V3 was a single-seed quick run on an NVIDIA H100 with `torch.compile`, reported as directional only.

Metrics. We distinguish a *score* (a cosine similarity over a single pair of embeddings) from a *metric* (an aggregation over the score population). For closed-set retrieval we report Precision@1 (P@1) and Precision@5 (P@5): P@5 is the fraction of the top-5 retrieved gallery epochs (excluding the query itself) that share the query identity, averaged across queries. For signal quality we report SI-SNR in decibels. For closed-set pairwise verification we report AUROC and equal-error rate (EER): AUROC is the area under the receiver-operating-characteristic curve constructed by sweeping a threshold over the cosine-similarity score, and EER is the threshold at which false-accept and false-reject rates coincide. Both are computed over all test-pool embedding pairs (Table III); held-out-subject open-set scoring is described separately in the protocol paragraph.

V. Results and Discussion

A. Cross-Version Best P@1

Table I summarizes the best Precision@1 of each version, selected per noise family across the three metric heads. The proposed V4 model is the strongest configuration on every noise type, with the clearest margins on power-line and EMG. V3 reaches a competitive power-line score (its tuned Mamba preset narrows the spectral residual the 50 Hz line leaves behind), but its Gaussian and EMG numbers trail V1, V2, and V4: with a single seed and a different hardware target (H100 + `torch.compile`) the run is not directly comparable to the three-seed RTX 5090 numbers and is reported only as a directional reference.

B. Detailed V4 Results

Table III reports the full V4 metric grid for the three metric heads. ResNet-34 with ArcFace is consistently the strongest head in P@1, while ResNet-18 with multi-similarity is competitive on power-line. The denoiser reaches its highest SI-SNR

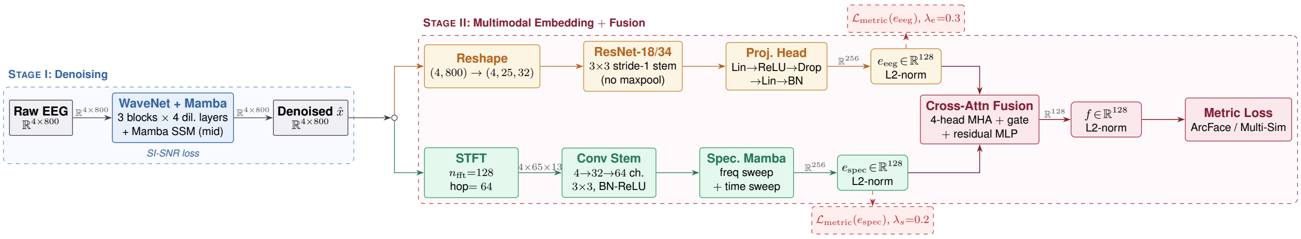

Fig. 1. V4 multimodal architecture. The WaveNet+Mamba denoiser feeds two parallel branches: a ResNet metric embedder on the denoised waveform and a spectrogram Mamba branch with frequency- and time-axis sweeps. A gated self-attention fusion module fuses the two L2-normalized embeddings into a single 128-d biometric embedding. Auxiliary metric losses on each branch (dashed) prevent fusion collapse during joint training.

TABLE I
BEST PRECISION@1 (P@1) PER VERSION AND NOISE TYPE. HIGHER IS BETTER. V3 IS A SINGLE-SEED QUICK RUN.

Noise	V1	V2	V3†	V4
Gaussian	0.822	0.798	0.749	**0.824**
Power-line	0.860	0.858	0.869	**0.877**
EMG	0.824	0.811	0.758	**0.838**

† Single-seed H100 quick run; reported for reference only.

TABLE II
CITED PRIOR-WORK NUMBERS ON EEG-BASED IDENTIFICATION. EACH ROW USES ITS SOURCE PAPER'S PROTOCOL; NO BASELINE IS RE-IMPLEMENTED UNDER THE AEP-HYBRID PROTOCOL OF THIS PAPER. V4 RESULTS ARE IN TABLE I.

Method	Dataset	Subj.	Metric	Score
MindID [4][1]	EEG-ID	8	Acc.	0.984
EEG-Triplet [3][2]	PhysioNet MI	109	Acc.	0.961
BrainNet [2][3]	ERP-CORE	40	EER	0.027
AISEI v0 [1][4]	PhysioNet AEP	20	P@1	0.860

[1] Within-session resting-state EEG; attention-RNN; no synthetic noise injection.
[2] Motor-imagery task; triplet loss on raw signals; session-mixed split.
[3] Siamese network on ERP epochs; cross-session evaluation; open-set verification on enrolled subjects only.
[4] Same dataset and same AEP-Hybrid protocol as this work; WaveNet-only denoiser; power-line noise.

in the power-line condition (the 50 Hz sinusoid is structurally easy to remove); Gaussian and EMG SI-SNR are lower but still in the regime expected for non-stationary residuals.

C. Ablation: Mamba and Multimodal Fusion

Table IV isolates the contribution of each architectural ingredient on the strongest head (ResNet-34 + ArcFace) using P@1 as the discriminating metric. Inserting a single Mamba block into the WaveNet denoiser (V1→V2) leaves P@1 essentially unchanged and even slightly hurts it, suggesting that swapping a single fixed-receptive-field convolutional layer for a state-space block does not, on its own, re-shape the embedding geometry. Adding the spectrogram branch and gated self-attention fusion (V2→V4) is what unlocks retrieval gains of +2.6, +1.9, and +2.7 percentage points respectively over V2, and +0.2, +1.7, and +1.4 over V1. We do not attempt to attribute SI-SNR deltas to the Stage II additions: each version retrains its own Stage I denoiser independently, so

TABLE III
V4 MULTIMODAL RESULTS, MEAN±STD OVER THREE SEEDS. THE RESNET-34+ARCFACE HEAD IS THE STRONGEST BY P@1 ON ALL NOISE FAMILIES.

Noise	Head	P@1	P@5	SI-SNR	AUROC	EER
Gauss.	R34+MS	.764±.050	.743±.053	12.15±.26	.443±.021	.376±.072
	R18+MS	.782±.028	.755±.035	12.15±.26	.441±.036	.369±.046
	R34+Arc	**.824±.036**	**.803±.040**	12.15±.26	.479±.095	.356±.059
P-line	R34+MS	.819±.040	.794±.049	32.38±1.06	.533±.051	.408±.024
	R18+MS	.831±.047	.806±.052	32.38±1.06	.450±.015	.371±.021
	R34+Arc	**.877±.006**	**.855±.005**	32.38±1.06	.501±.044	.360±.039
EMG	R34+MS	.764±.080	.747±.075	13.83±.35	.439±.065	.389±.039
	R18+MS	.793±.067	.775±.069	13.83±.35	.422±.038	.356±.066
	R34+Arc	**.838±.057**	**.827±.065**	13.83±.35	.514±.082	.350±.058

MS: multi-similarity loss; Arc: ArcFace; R18/R34: ResNet-18/34.

TABLE IV
ARCHITECTURAL ABLATION ON RESNET-34+ARCFACE BY P@1. Δ ROWS ARE V4 MINUS V1. STAGE I IS RETRAINED INDEPENDENTLY PER VERSION, SO SI-SNR IS NOT USED HERE AS AN ABLATION METRIC.

Noise	Variant	P@1
Gaussian	V1: WaveNet	0.822
	V2: + Mamba	0.798
	V4: + Spec + Fusion	**0.824**
	Δ V4−V1	+0.002
Power-line	V1: WaveNet	0.860
	V2: + Mamba	0.858
	V4: + Spec + Fusion	**0.877**
	Δ V4−V1	+0.017
EMG	V1: WaveNet	0.824
	V2: + Mamba	0.811
	V4: + Spec + Fusion	**0.838**
	Δ V4−V1	+0.014

cross-version SI-SNR differences reflect independent training runs of nominally the same architecture rather than any contribution from the spectrogram branch or fusion module. Per-version Stage I SI-SNR (mean across seeds, ResNet-34+ArcFace configuration) is V1: 10.64/19.98/11.65 dB; V2: 10.65/19.61/11.67 dB; V4: 12.15/32.38/13.83 dB on Gaussian / power-line / EMG, and is intended only as descriptive Stage I quality.

D. Discussion

Where multimodal helps. The largest mean-P@1 jumps occur on power-line noise, consistent with the spectrogram localizing the 50 Hz line and the gate suppressing it before fusion. EMG is intermediate (broadband but high-frequency-tilted); Gaussian gains are smallest, consistent with white noise being structureless in both views.

Effect sizes and the Mamba-alone variant. V4-over-V1 and V2-vs-V1 mean-P@1 deltas both fall within across-seed standard deviation, so we report them as magnitude observations rather than tested claims (see Limitations). The implication is that the gains should be read as a property of the full multimodal stack — denoiser, spectrogram branch, and gated fusion together — rather than an isolated Stage-I Mamba effect.

Verification gap. Held-out-subject AUROC of the centroid min-distance score stays close to chance (≈ 0.50) across all V4 configurations, confirming that threshold-free verification of unseen identities remains the open problem. The AUROC (0.42–0.53) and EER (0.35–0.41) columns in Table III are computed on the closed-set test pool (session- and epoch-disjoint, all subjects seen during training) and indicate within-subject session variability, not novel-identity verification; calibration is left to future work.

Limitations and future work. Each limitation below is paired with the planned remedy. (i) The dataset is small (20 subjects, 4 channels), so absolute deltas are within across-seed noise; we plan a multi-cohort recording with ≥ 100 subjects to support paired significance at the per-noise level. (ii) Noise is synthetic by controlled injection; we plan a live-environment recording session with hardware artefacts (movement, blink, electrode drift) to retire the synthetic-only caveat. (iii) Held-out-subject AUROC is near chance; we plan verification calibration via Platt scaling on a held-out validation cohort and a per-subject score-normalisation study. (iv) Same-protocol re-implementation of external EEG-biometric baselines (EEGNet, ChronoNet, ICA+SVM) and a cross-dataset transfer evaluation are out of scope here and are scheduled for the journal version of this work; Table II collects cited numbers under their own protocols as an informational, not controlled, positioning of the V4 result.

VI. Conclusion

We presented a multimodal Mamba-augmented framework for noise-robust EEG biometric identification extending [1]: V4 combines a WaveNet+Mamba denoiser, a spectrogram Mamba branch, and a gated self-attention fusion module trained with auxiliary branch losses. Under the AEP-Hybrid protocol (Sec. IV) on the PhysioNet AEP dataset, V4 attains the highest mean Precision@1 on all three noise families (82.4%/87.7%/83.8%); held-out-subject AUROC ≈ 0.50 motivates calibration as future work.

References

[1] L. N. Vu, B. T. N. Tran, H. C. Bang, and N. M. Hanh, "Biometric identification through auditory EEG signatures," in *Proc. International Conference on Artificial Intelligence for Sustainable Engineering and Innovation (AISEI)*, accepted, in press, Irbid, Jordan, 2026.

[2] M. Fallahi, T. Strufe, and P. Arias-Cabarcos, "BrainNet: Improving brainwave-based biometric recognition with siamese networks," in *Proc. IEEE International Conference on Pervasive Computing and Communications (PerCom)*, 2023, pp. 53–60.

[3] H. Yang, W. Zheng, Y. Zhang, and B.-L. Lu, "EEG authentication based on deep learning of triplet loss," *Neural Network World*, vol. 32, no. 5, pp. 269–283, 2022.

[4] X. Zhang, L. Yao, S. S. Kanhere, Y. Liu, T. Gu, and K. Chen, *MindID: Person identification from brain waves through attention-based recurrent neural network*, 2017. arXiv: 1711.06149 [cs.LG].

[5] N. A. Alzahab et al., "Auditory evoked potential electroencephalography-biometric dataset," *Data in Brief*, vol. 57, p. 111 065, 2024.

[6] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 4690–4699.

[7] X. Wang, X. Han, W. Huang, D. Dong, and M. R. Scott, "Multi-similarity loss with general pair weighting for deep metric learning," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 5022–5030.

[8] A. Gu and T. Dao, *Mamba: Linear-time sequence modeling with selective state spaces*, 2023. arXiv: 2312.00752 [cs.LG].

[9] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," in *International Conference on Learning Representations (ICLR)*, 2022.

[10] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: A compact convolutional neural network for EEG-based brain-computer interfaces," *Journal of Neural Engineering*, vol. 15, no. 5, p. 056 013, 2018.

[11] R. T. Schirrmeister et al., "Deep learning with convolutional neural networks for EEG decoding and visualization," *Human Brain Mapping*, vol. 38, no. 11, pp. 5391–5420, 2017.

[12] A. van den Oord et al., *WaveNet: A generative model for raw audio*, 2016. arXiv: 1609.03499 [cs.SD].

[13] Y. Luo and N. Mesgarani, "Conv-TasNet: Surpassing ideal time-frequency magnitude masking for speech separation," *IEEE/ACM Trans. Audio, Speech, and Language Processing*, vol. 27, no. 8, pp. 1256–1266, 2019.

[14] A. Vaswani et al., "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 5998–6008.

[15] Y. Wang et al., "EEG-Mamba: A selective state-space model for cross-subject EEG decoding," in *Proc. IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 2024.

[16] R. Hadsell, S. Chopra, and Y. LeCun, "Dimensionality reduction by learning an invariant mapping," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2006, pp. 1735–1742.

[17] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 815–823.

[18] Y. Gong, Y.-A. Chung, and J. Glass, "AST: Audio spectrogram transformer," in *Proc. Interspeech*, 2021.

[19] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[20] K. Musgrave, S. Belongie, and S.-N. Lim, "PyTorch metric learning," in *arXiv:2008.09164*, 2020.